

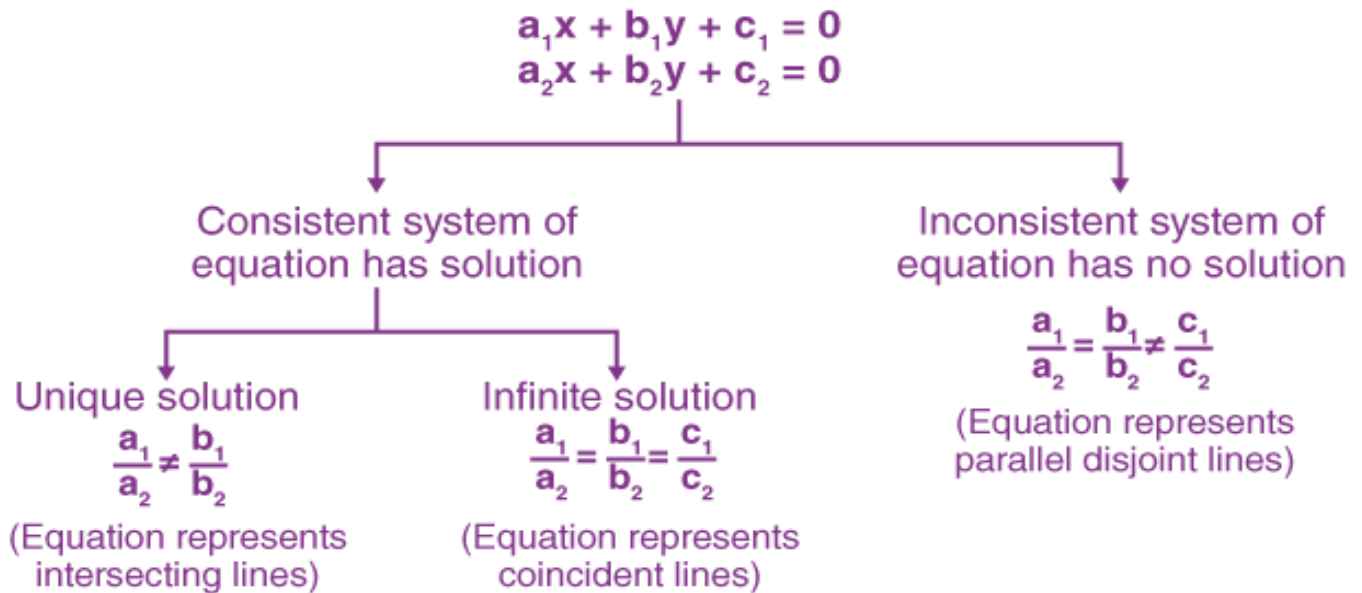
Geometrical Representation of Linear System in two variables

Inconsistent Linear System:

If the **linear system** has **no solution**, then it is said to be an **inconsistent linear system**.

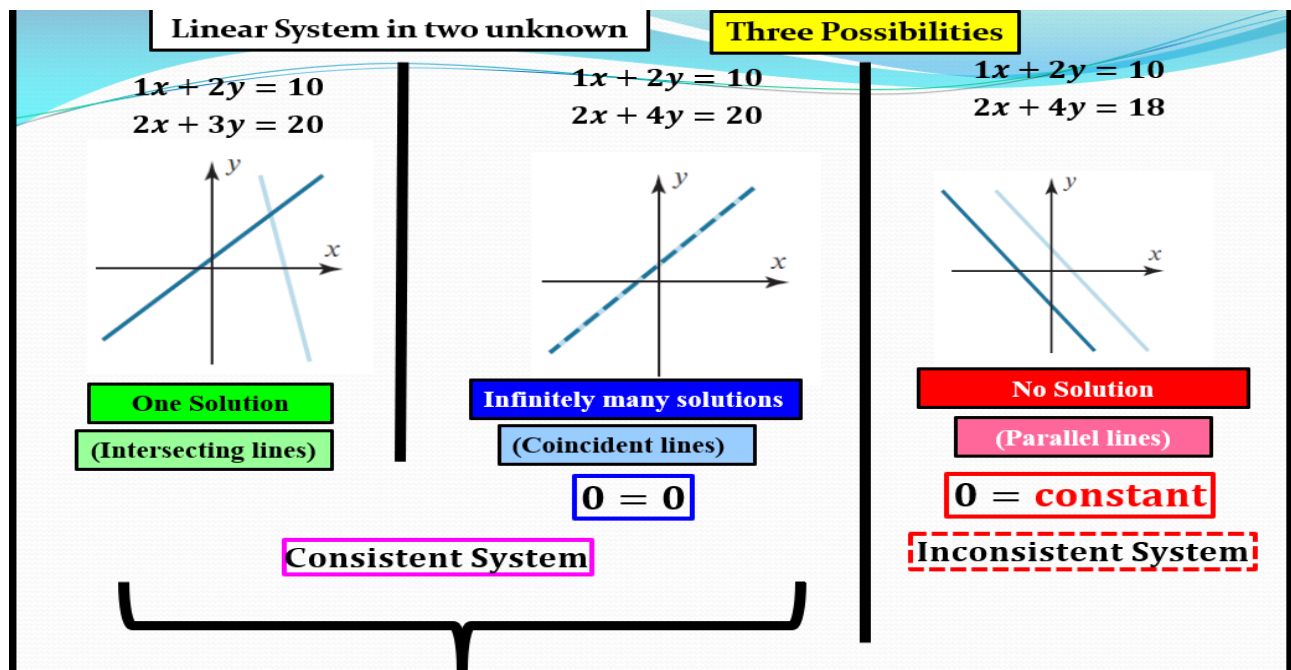
Consistent Linear System:

If the **linear system** has a **solution**, then it is said to be a **consistent linear system**.



Note:

- A **consistent linear system** of **two equations** in **two unknowns** has **either one solution** or **infinitely many solutions** -- there are no other possibilities.



EXAMPLE 1

A Linear System with No Solutions

Solve the linear system

$$\begin{cases} x + y = 4 \\ 3x + 3y = 6 \end{cases}$$

Solution: Check yourself ----- $0 = 6$.

This is **not possible**, so the given system has **no solution**. Hence, it is an **inconsistent linear system**.

Geometrically, this means that **the lines corresponding to the equations in the original system are parallel and distinct**.

EXAMPLE 2

A Linear System with One Solution

Solve the linear system

$$\begin{cases} x - y = 1 \\ 2x + y = 6 \end{cases}$$

Solution: Verify that the system has the **one solution** (unique solution) $x = \frac{7}{3}$, $y = \frac{4}{3}$. So, $(x, y) = \left(\frac{7}{3}, \frac{4}{3}\right)$. Hence, the given system is a **consistent linear system**.

Geometrically, this means that **the lines corresponding to the equations in the original system are intersecting at a single point**.

EXAMPLE 3

A Linear System with Infinitely many Solutions

Solve the linear system

$$\begin{cases} 4x - 2y = 1 \\ 16x - 8y = 4 \end{cases}$$

Solution: The solutions of the system are those values of x and y that satisfy the single equation. Let $y = t$, so

$$x = \frac{1}{4} + \frac{1}{2}t$$

Hence, $(x, y) = \left(\frac{1}{4} + \frac{1}{2}t, t\right)$, $t \in \mathbb{R}$ is solution of given system. Hence, the given system is a **consistent linear system**, since it has **infinitely many solutions**.

Geometrically, this means that **the lines corresponding to the equations in the original system are coincident lines.**

Note:

We can obtain specific numerical solutions for different values of t . If $t = 0$, yields the solution $\left(\frac{1}{4}, 0\right)$. Similarly, $t = 1$ yields the solution $\left(\frac{3}{4}, 1\right)$.

Note:

Equally-Determined System:

If

No. of Equations = No. of variables or Rank of the matrix

then the system is known as **Equally-determined** system.

Example:

Consider the system below

$$\begin{cases} a_1x + b_1y = d_1 \\ a_2x + b_2y = d_2 \end{cases}$$

In the given system, we can observe that we have two equations with two variables x and y .

Under-Determined System:

If

No. of Equations < No. of variables or Rank of the matrix

then the system is known as **under-determined** system.

Example:

Consider the system below

$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \end{cases}$$

In the given system, we can observe that we have two equations with three variables x, y and z .

Over-Determined System:

If

No. of Equations $>$ No. of variables or Rank of the matrix

then the system is known as **over-determined** system.

Example:

Consider the system below

$$\begin{cases} a_1x + b_1y = d_1 \\ a_1x + b_1y = d_2 \\ a_2x + b_2y = d_2 \end{cases}$$

In the given system, we can observe that we have two equations with three variables x, y and z .

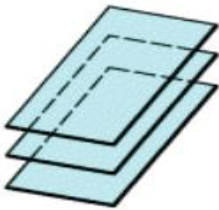
Geometrical Representation of Linear System in three variables

Consider a linear system of three equations in three unknowns

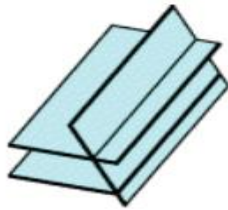
$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$$

in which the graphs of the equations are planes. The **solutions of the system**, if any, **correspond** to points **where all three planes intersect**, so again we see that there are only three possibilities.

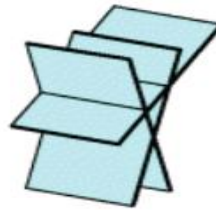
- 1) **No solution**
- 2) **One solution**
- 3) **Infinitely many solutions**



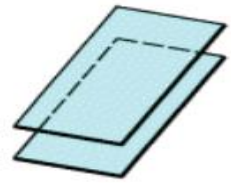
No solutions
(three parallel planes;
no common intersection)



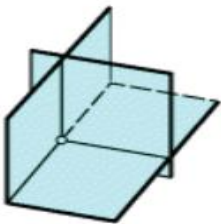
No solutions
(two parallel planes;
no common intersection)



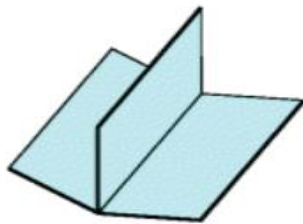
No solutions
(no common intersection)



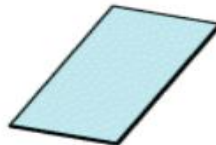
No solutions
(two coincident planes
parallel to the third;
no common intersection)



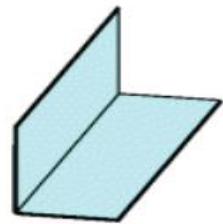
One solution
(intersection is a point)



Infinitely many solutions
(intersection is a line)



Infinitely many solutions
(planes are all coincident;
intersection is a plane)



Infinitely many solutions
(two coincident planes;
intersection is a line)

Important Note:

Every system of linear equations has **no solution**, **one solution**, or **infinitely many solutions**. There are no other possibilities.

EXAMPLE 4 Solve the linear system

$$\begin{cases} x + y + 2z = 9 & \text{--- (1)} \\ 2x + 4y - 3z = 1 & \text{--- (2)} \\ 3x + 6y - 5z = 0 & \text{--- (3)} \end{cases}$$

Solution:

Using equation (1) and equation (2)

$$\begin{array}{r} 2x + 2y + 4z = 18 \\ + 2x + 4y - 3z = 1 \\ \hline -2y + 7z = 17 \quad \rightarrow (4) \end{array}$$

Using equation (1) and equation (3)

$$\begin{array}{r} 3x + 3y + 6z = 27 \\ + 3x + 6y - 5z = 0 \\ \hline -3y + 11z = 27 \quad \rightarrow (5) \end{array}$$

Using equation (4) and equation (5)

$$\begin{array}{r} -6y + 21z = 51 \\ -6y + 22z = 54 \\ \hline -z = -3 \\ z = 3 \end{array}$$

Put $z = 3$ in equation (4)

$$\begin{array}{r} -2y + 7(3) = 17 \\ -2y + 21 = 17 \\ y = 2 \end{array}$$

Put $y = 2$ and $z = 3$ in equation (1).

$$\begin{array}{r} x = 9 - 2 - 2(3) \\ x = 1 \end{array}$$

So, the solution of given system is $(x, y, z) = (1, 2, 3)$, which is unique. So, it has **one solution**. **Geometrically**, it represents that the three planes intersect at a **unique point**.

EXAMPLE 5

Solve the linear system

$$\begin{cases} x - y + 2z = 5 \\ 2x - 2y + 4z = 10 \\ 3x - 3y + 6z = 15 \end{cases}$$

Solution:

This system can be solved by inspection, since the second and third equations are multiples of the first. Geometrically, this means that the three planes coincide and that those values of x , y , and z that satisfy the equation

$$x - y + 2z = 5 \text{ --- (1)}$$

Automatically satisfy all three equations. Thus, it suffices to find the solutions of equation (1). We can do this by first solving equation (1) for x in terms of y and z , then assigning arbitrary values s and t (parameters) to these two variables

$$y = t, \quad z = s$$

and then expressing the solution by the three parametric equations implies that

$$x = 5 + t - 2s$$

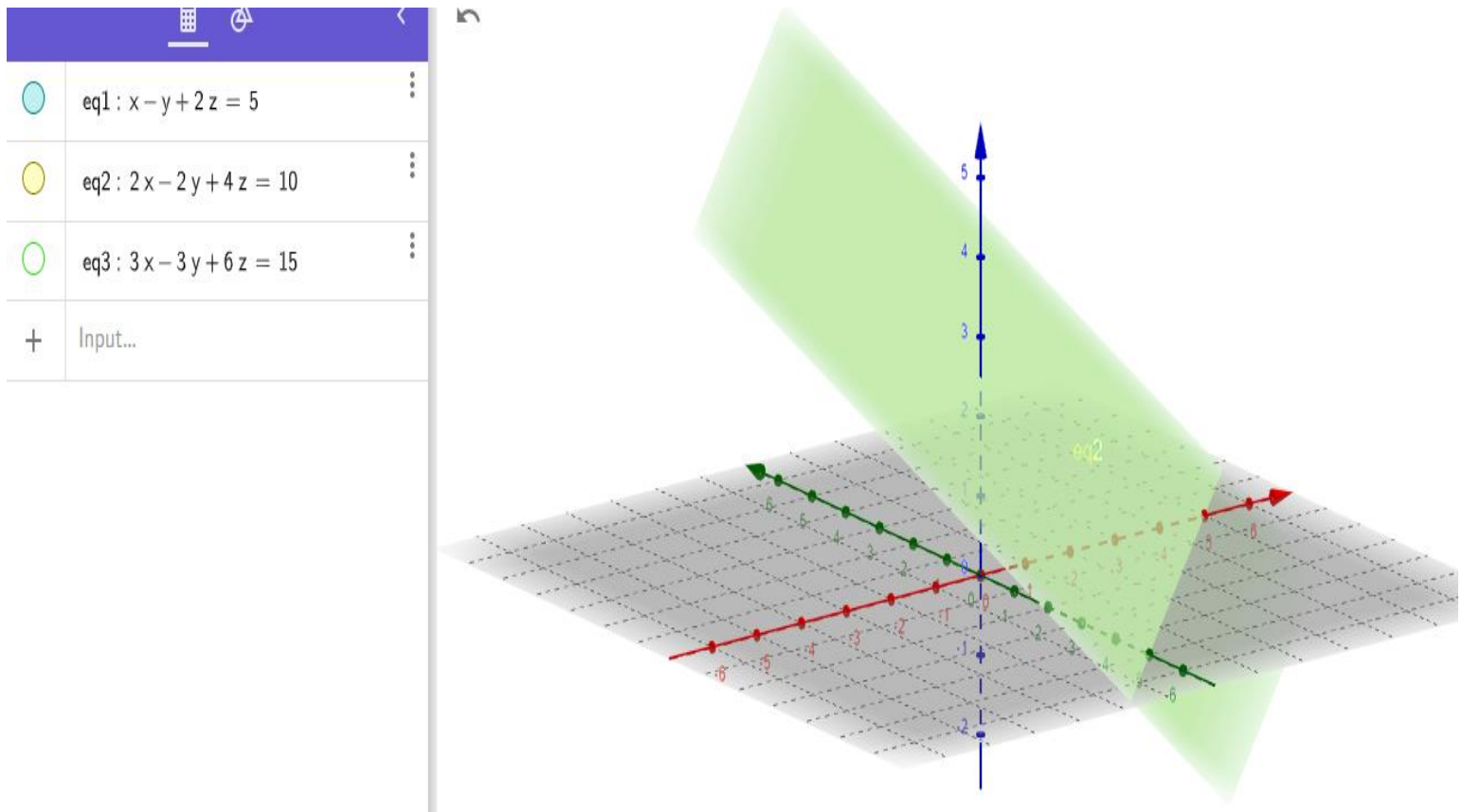
So $(x, y, z) = (5 + t - 2s, t, s)$ is the solution of the above consistent linear system.

Note:

Specific solutions can be obtained by choosing numerical values for the parameters t and s . For example, taking $t = 1$ and $s = 0$ yields the solution $(x, y, z) = (6, 1, 0)$.

Use of GeoGebra Software for visualization of solutions and geometry

To open **GeoGebra Software** online, [Click Here](#)



Work to do

Q. Solve the following linear system of equations:

$$\begin{cases} 2x + 4y + 6z = -12 \\ 2x - 3y - 4z = 15 \\ 3x + 4y + 5z = -8 \end{cases}$$

Exercise 1.1 [*Elementary Linear Algebra with Applications by Howard Anton*]

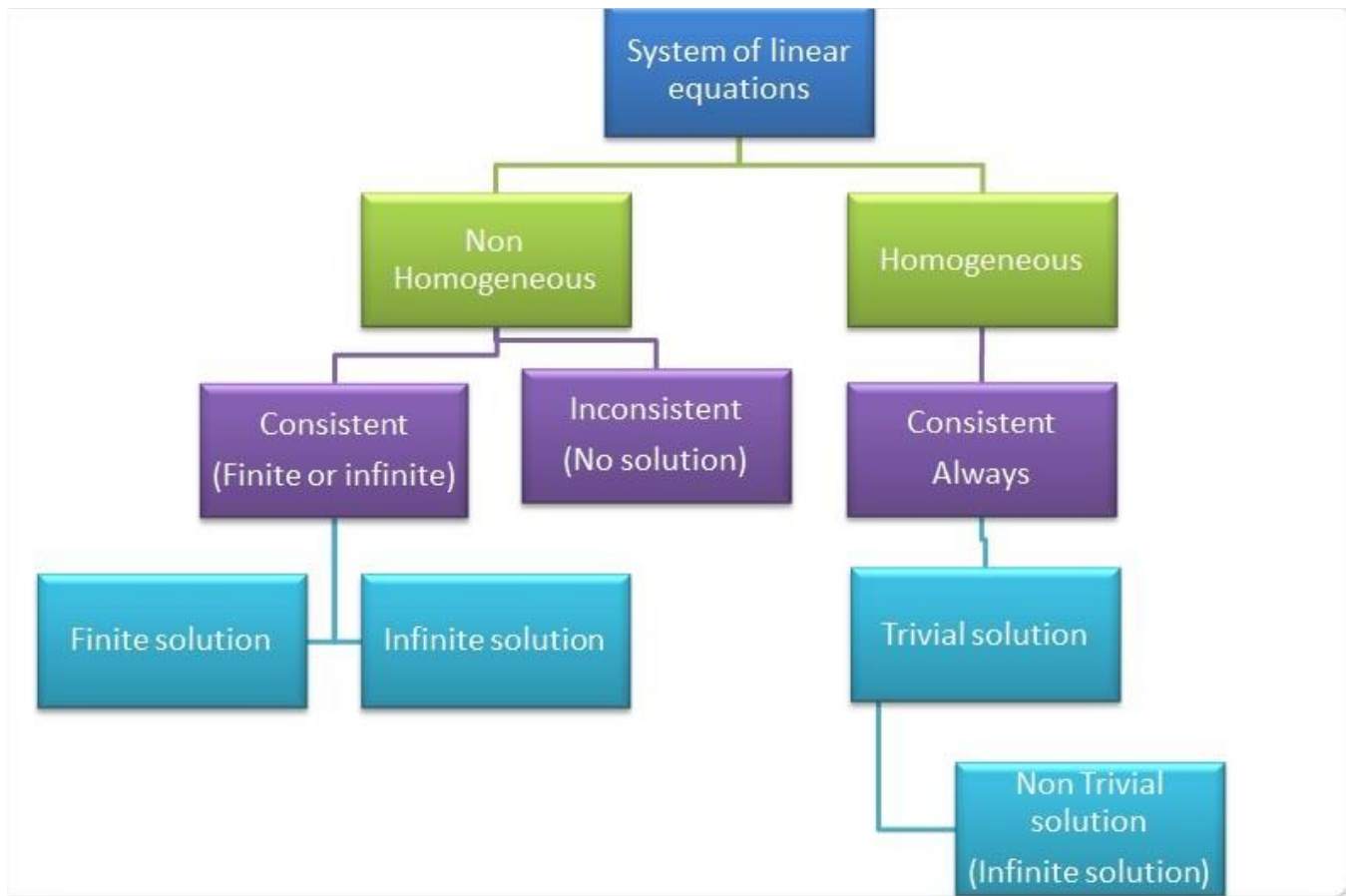
Questions 6-14

9. In each part, find the solution set of the linear equation by using parameters as necessary.
- a) $7x - 5y = 3$
 - b) $-8x_1 + 2x_2 - 5x_3 + 6x_4 = 1$

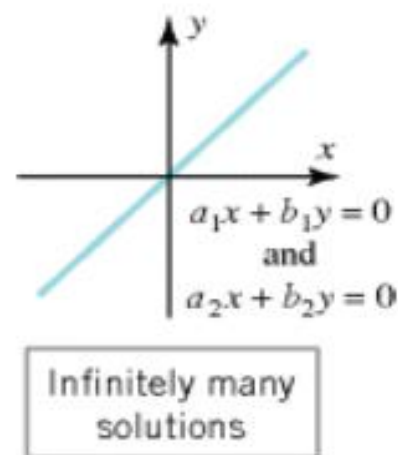
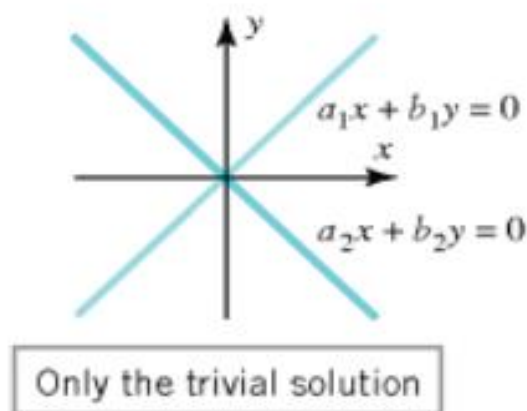
Homogeneous Linear System

A system of linear equations is said to be a **homogeneous linear system**, if its constant term is equal to zero. **For example**

$$\begin{cases} 3x + 4y = 0 \\ -2x + 5y = 0 \end{cases}$$



- ❖ **Every system of linear equations is consistent** because all such systems have $x = 0, y = 0$ as a solution. This solution is called **trivial solution**.
- ❖ If there are other solutions, they are called **non trivial solutions** (infinite many solutions).
- ❖ A **homogenous system always has the trivial solution**. There are only two possibilities for its solutions:



System of Linear Equation

Homogeneous

Always
consistent

$$AX = 0$$

Trivial
solution

Zero
solution

$\rho(A) = \text{no of variable}$
or
 $|A| \neq 0$
or
Non-singular

Non-Trivial
solution

Infinite
solution

Non-Zero
solution

$\rho(A) < \text{no of variable}$
or
 $|A| = 0$
or
Singular matrix

For a homogeneous system of linear equations, having non-trivial solution:

- The **rank of the matrix** $\rho(A)$ should be **less than the number of variables**. **OR**
- The **determinant of a matrix** should be **equal to zero**.

Practice Questions for Students

Q. In each part, find the augmented matrix for the given system of linear equations.

a)
$$\begin{cases} 3x_1 - 2x_2 = -1 \\ 4x_1 + 5x_2 = 3 \\ 7x_1 + 3x_2 = 2 \end{cases}$$

b)
$$\begin{cases} 2x_1 + 2x_3 = 1 \\ 3x_1 - x_2 + 4x_3 = 7 \\ 6x_1 + x_2 - x_3 = 0 \end{cases}$$

c)
$$\begin{cases} x_1 + 2x_2 - x_4 + x_5 = 1 \\ 3x_2 + x_3 - x_5 = 2 \\ x_3 + 7x_4 = 1 \end{cases}$$

d)
$$\begin{cases} x_1 = 1 \\ x_2 = 2 \\ x_3 = 3 \end{cases}$$

Q. In parts (I)–(VIII), determine whether the statement is true or false, justify your answer.

- A linear system whose equations are all homogeneous must be consistent.
- Multiplying a linear equation through by zero is an acceptable elementary row operation.
- The linear system

$$\begin{cases} x - y = 3 \\ 2x - 2y = k \end{cases}$$

cannot have a unique solution, regardless of the value of k .

- A single linear equation with two or more unknowns must always have infinitely many solutions.

- V. If the number of equations in a linear system exceeds the number of unknowns, then the system must be inconsistent.
- VI. If each equation in a consistent linear system is multiplied through by a constant c , then all solutions to the new system can be obtained by multiplying solutions from the original system by c .
- VII. Elementary row operations permit one equation in a linear system to be subtracted from another.
- VIII. The linear system with corresponding augmented matrix $\left(\begin{array}{cc|c} 2 & -1 & 4 \\ 0 & 0 & -1 \end{array}\right)$ is consistent.